0.1 The full integral symplectic targets and their retained products

Statement E.56 (Proved). Exact transvection targets at every modulus prime to six. *Let J be the alternating chain matrix of E.52.7, put $\Omega = J^{-1}$, and write*

$$a = e_1, \qquad b = e_2, \qquad c = e_1 + e_3, \qquad d = e_4, \qquad W = \langle a, b \rangle. \tag{E.56.1}$$

For $T = \mathbf{Z}/m$, where $(m, 6) = 1$, or $T = R = \prod_{p \geq 5} \mathbf{Z}_p$, define

$$\begin{aligned} T_v(t) &= I + tvv^{\mathrm{t}}J, \\ \mathcal{P}(T) &= \mathrm{Stab}_{\mathrm{Sp}_4(T,J)}(c), \\ \mathcal{P}^{\mathrm{rt}}(T) &= J\mathcal{P}(T)J^{-1} \subset \mathrm{Sp}_4(T, \Omega). \end{aligned} \tag{E.56.2}$$

The four original column transvections $T_a(1), T_b(1), T_{c-a}(1), T_d(1)$ generate $\mathrm{Sp}_4(T, J)$; the first three generate exactly $\mathcal{P}(T)$. For $T = R$, generation means closed topological generation. The exact structure of the latter group is

$$\begin{aligned} \mathcal{P}(T) &= U(T) \rtimes \mathrm{SL}_2(T), \\ U(T) &= \{n(w, z) : w \in W(T),\ z \in T\}, \\ n(w, z) &= I + wc^{\mathrm{t}}J + cw^{\mathrm{t}}J + zcc^{\mathrm{t}}J, \\ n(w, z)n(w', z') &= n(w + w', z + z' + \langle w, w' \rangle). \end{aligned} \tag{E.56.3}$$

Here $\langle u, v \rangle = u^{\mathrm{t}}Jv$, and the displayed SL_2 acts on W and fixes c, d individually.

In the literal right convention of E.52, the generators are $M_i = T_{e_i}(1)^{\mathrm{t}}$. Their full image is $\mathrm{Sp}_4(T, \Omega)$ and their first-three image is exactly $\mathcal{P}^{\mathrm{rt}}(T)$. Those first three matrices fix the row c^{t}, or equivalently the column $Jc = -e_4$. The precise source comparison is

$$J^{-1}M_iJ = T_{e_i}(-1), \qquad \mathrm{Ad}(J^{-1}) \circ q^{\mathrm{rt}}_{T,n} = q^{\mathrm{col}}_{T,n} \circ \iota, \quad \iota(\sigma_i) = \sigma_i^{-1}, \quad n = 4, 5. \tag{E.56.4}$$

The map ι is the actual involutive braid automorphism given on all the displayed generators.

Retain the literal Ferrari maps to T_L, T_{23} and the exact quotient S_L from E.38–E.43, and the four- and five-strand Hurwitz maps to A_{96} from E.46. With their specified original generator maps, the actual four-strand joint image is

$$(T_{23} \times_{S_L} T_L) \times \mathcal{P}^{\mathrm{rt}}(T) \times A_{96}, \tag{E.56.5}$$

and the actual five-strand joint image is

$$\mathrm{Sp}_4(T, \Omega) \times A_{96}. \tag{E.56.6}$$

In particular the actual joint image of the two four-strand maps to T_L and the new symplectic target is $T_L \times \mathcal{P}^{\mathrm{rt}}(T)$. All these statements retain their reductions and the actual inverse-limit maps. The corrected words of E.55 generate the same displayed joint targets.

Proof. We give elementary matrix elimination and then determine the common quotients through the specified braid maps.

The exact conventions and the Heisenberg subgroup. The integral matrix with columns a, b, c, d has determinant one and satisfies

$$(a\ b\ c\ d)^{\mathrm{t}}J(a\ b\ c\ d) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{E.56.7}$$

Thus W and $\langle c, d\rangle$ are orthogonal symplectic planes. Direct multiplication gives $T_v(t)T_v(u) = T_v(t+u)$ and $gT_v(t)g^{-1} = T_{gv}(t)$ for a symplectic g. The first two transvections give the upper and lower elementary matrices on W. For a unit u, put

$$w_W(u) = T_a(u)T_b(u^{-1})T_a(u). \tag{E.56.8}$$

Its matrix on W is $\left(\begin{smallmatrix} 0 & u \\ -u^{-1} & 0 \end{smallmatrix}\right)$, and $w_W(u)w_W(-1) = \mathrm{diag}(u, u^{-1})$.

The half turn $h = w_W(1)^2$ sends $c - a$ to $c + a$ and fixes c, d. Orthogonality of a, c gives the exact polynomial identities

$$\begin{aligned} T_{c+a}(t)T_{c-a}(t) &= T_c(2t)T_a(2t), \\ T_{c+a}(t)T_a(-t)T_c(-t) &= I + t\big(ac^{\mathsf{t}}J + ca^{\mathsf{t}}J\big). \end{aligned} \tag{E.56.9}$$

Consequently the first three original transvections give $T_c(t)$ and $n(ta, 0)$ for every parameter t. Conjugation by $w_W(1)$ gives $n(tb, 0)$ as well. For finite T, every parameter is an integer power of the corresponding unit-parameter transvection; division by two in the first formula is legitimate because two is a unit. Formula E.56.3 follows by multiplication, and in particular

$$\begin{aligned} n(xa + yb, z) &= n(xa, 0)n(yb, 0)T_c(z - xy), \\ [n(xa, 0), n(yb, 0)] &= T_c(2xy). \end{aligned} \tag{E.56.10}$$

These are actual words in the original three generators, with every parameter and central term retained.

To identify the whole stabilizer, a symplectic matrix fixing c induces a determinant-one matrix A on $c^{\perp}/Tc \simeq W$. Multiplying by the inverse of the displayed lift of A makes that induced action the identity. The resulting matrix sends

$$c \longmapsto c, \qquad d \longmapsto d + w + zc, \qquad v \longmapsto v + \langle w, v\rangle c \quad (v \in W). \tag{E.56.11}$$

The last formula is forced by pairing its image with the image of d. Hence it is precisely $n(w, z)$, uniquely. This proves E.56.3 through the indicated matrix map.

Over a local ring $\mathbf{Z}/p^r$, every $A = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$ of determinant one has a unit in its first column. When α is a unit, the exact factorization is

$$A = E_{21}(\gamma/\alpha)\,\mathrm{diag}(\alpha, \alpha^{-1})E_{12}(\beta/\alpha). \tag{E.56.12}$$

Otherwise left multiplication by $w_W(1)$ makes the top entry of that column a unit. This proves elementary generation of the required SL_2 and, with E.56.10, proves that the first three original generators give the full stabilizer over each local ring.

Eliminating a primitive vector. Adding $T_d(1)$ gives the same elementary SL_2 on $\langle c, d\rangle$. We describe an actual word taking any primitive vector $v = v_W + zc + td$ to c over $\mathbf{Z}/p^r$. If neither z nor t is a unit, a coordinate of v_W is a unit. Choose $w = a$ or b so that $\langle w, v_W\rangle$ is a unit. Multiplication by $n(w, 0)$ changes the c-coordinate to $z + \langle w, v_W\rangle$, which is then a unit. If the d-coordinate is not yet a unit, the plane rotation $w_{\langle c,d\rangle}(1)$ makes it one. Rename the resulting coordinates v_W, z, t, with t a unit.

Now apply, in this order,

$$n(-t^{-1}v_W, 0), \quad T_c(-z/t), \quad \mathrm{diag}(t, t^{-1})\big|_{\langle c,d\rangle}, \quad w_{\langle c,d\rangle}(1). \tag{E.56.13}$$

The first map removes v_W and leaves z unchanged, because $\langle -t^{-1}v_W, v_W\rangle = 0$. The next removes z; the last two send td to d and then to c. Every factor has already been expressed as a word in the four original transvections.

For an arbitrary symplectic g, apply this word to gc. The resulting product fixes c, so it belongs to the explicitly generated stabilizer. This proves generation of the entire symplectic group over each local ring. For a general m, CRT parameters supported at a single prime-power factor make each original transvection the identity in all other factors. The local words therefore give the whole product over $\mathbf{Z}/m$. The stabilizer proof has the same property. Compactness and the cofinal divisibility system of moduli give the asserted closed images over R.

The transpose convention is also literal. Since $J^{\mathsf{t}} = -J$, multiplication gives $T_{e_i}(1)^{\mathsf{t}} = JT_{e_i}(-1)J^{-1}$. The inverse-generator substitution preserves both the Artin braid relations and the distant commutation relations, and its square is the identity. It is therefore the automorphism ι in E.56.4. The equalities on every braid generator prove that displayed morphism. They also prove both stated fixed-vector descriptions, including $Jc = -e_4$. The saved level-five and level-seven permutation maps are compared with these transposes on every nonzero vector in the certificate below.

Perfectness and the actual simple quotients. The groups just generated are perfect at every finite level. On a symplectic plane, conjugation by $\mathrm{diag}(2, 1/2)$ multiplies the upper elementary parameter by four; its commutator with $E_{12}(t)$ is $E_{12}(3t)$. The lower parameter is multiplied by $1/4$, and the corresponding difference $-3/4$ is a unit. Thus the plane SL_2 is perfect. Its commutators with $n(ta, 0)$ give all those elements, since the difference on a is one. Its difference on b is $-1/2$, also a unit, so the same holds for $n(tb, 0)$. Their commutators in E.56.10 give the whole central subgroup. Hence $\mathcal{P}(T)$ is perfect. For the full symplectic group, the primitive-vector elimination conjugates each original transvection to one in a symplectic plane; every generator is therefore a commutator. This proves perfectness of the full group too.

We include the finite-field simplicity argument needed at arbitrary primes. The projective action of $\mathrm{Sp}_4(\mathbf{F}_p, J)$ on lines has kernel exactly $\{\pm I\}$: a linear operator fixing all lines is scalar, and preservation of the alternating form gives these two scalars. The stabilizer of the line $\mathbf{F}_p c$ has the three orbits of sizes

$$1, \qquad p^2 + p, \qquad p^3. \tag{E.56.14}$$

For the perpendicular lines different from c, the plane SL_2 is transitive on the nonzero W component and $n(w, 0)$ changes the c component arbitrarily. For a line not perpendicular to c, scale its d coordinate to one; $n(w, z)$ then moves it to d. This proves the orbit assertion using the actual stabilizer already constructed.

A block containing the line of c is a union of these stabilizer orbits. The two intermediate possible sizes are $p^2 + p + 1$ and $p^3 + 1$. Neither divides the total number $N = p^3 + p^2 + p + 1$: one has $N = p(p^2 + p + 1) + 1$, and $p^3 + 1$ lies strictly between $N/2$ and N. The faithful projective action is consequently primitive. The subgroup $\{T_c(t) : t \in \mathbf{F}_p\}$ is abelian and normal in the line stabilizer, since replacing c by uc replaces t by $u^2 t$. Its conjugates generate the group by the proved transvection generation.

For completeness, the elementary normal-subgroup argument now proves simplicity. A nontrivial normal subgroup of a faithful primitive permutation group is transitive. Modulo such a subgroup, every conjugate of the displayed abelian subgroup has the same image: write each group element as a normal-subgroup element times an element of the point stabilizer, which normalizes that subgroup. The quotient is therefore generated by one abelian image. Perfectness makes it trivial. We have proved simplicity of $\mathrm{PSp}_4(\mathbf{F}_p)$ for every $p \geq 5$. Simplicity of $\mathrm{PSL}_2(\mathbf{F}_p)$ is proved by the same normal-subgroup argument in E.53.

The kernel of reduction from a matrix group modulo p^r to its reduction modulo p is a p-group: successive kernels embed, by $1 + p^j A \mapsto A$, into additive matrix spaces over $\mathbf{F}_p$.

The structure E.56.3 therefore shows that

$$\begin{aligned} \mathcal{P}(\mathbf{Z}/p^r) &\longrightarrow \mathrm{PSL}_2(\mathbf{F}_p), \\ \mathrm{Sp}_4(\mathbf{Z}/p^r) &\longrightarrow \mathrm{PSp}_4(\mathbf{F}_p) \end{aligned} \tag{E.56.15}$$

have solvable kernels. These are the actual maps induced on $c^{\perp}/Tc$, followed by reduction and the scalar quotient, in the first case, and the actual reduction and scalar quotient in the second.

Every proper normal subgroup of either source in E.56.15 is contained in the stated kernel. Indeed its image in the simple target is trivial or the whole target. In the latter case the resulting quotient is an image of the solvable kernel; it is also a quotient of a perfect group and hence must be trivial. Thus every nontrivial quotient retains the specified simple quotient in E.56.15. This proves the required assertion including all central and higher congruence kernels.

The complete target product with T_L. Consider the actual subdirect image of the two four-strand maps to T_L and $\mathcal{P}^{\mathrm{rt}}(\mathbf{Z}/m)$. A nontrivial Goursat common quotient is perfect, because the latter group is perfect. It has a nonabelian simple quotient. The matrix target is the product of its prime-power factors; their images in a simple group commute and are normal, so exactly one factor supplies that simple quotient. E.56.15 identifies it with the specified $\mathrm{PSL}_2(\mathbf{F}_p)$ map, up to an automorphism of that target.

For $p = 5, 7$, this would force that actual projective matrix map to factor through T_L. The complete literal products $T_L \times \mathcal{P}^{\mathrm{rt}}(\mathbf{F}_p)$ proved in E.50 and E.51 exclude that factorization. For $p \geq 11$, the order of $\mathrm{PSL}_2(\mathbf{F}_p)$ has a factor p, whereas the retained exact order of T_L has prime divisors only $2, 3, 5, 7$. This excludes every possible common quotient. Goursat therefore gives the whole actual product $T_L \times \mathcal{P}^{\mathrm{rt}}(\mathbf{Z}/m)$.

The projection $T_{23} \times_{S_L} T_L \to T_L$ has solvable kernel. In a common quotient with the perfect matrix target, quotienting by the image of that kernel gives a common quotient with T_L, just excluded. The original common quotient would then be solvable and perfect, and hence trivial. This proves the product with the whole old four-strand target.

The alternating factor and the full inverse limit. The simple group A_{96} is neither $\mathrm{PSL}_2(\mathbf{F}_p)$ nor $\mathrm{PSp}_4(\mathbf{F}_p)$ for any $p \geq 5$. The matrix group orders are

$$\frac{p(p^2-1)}{2}, \qquad \frac{p^4(p^2-1)(p^4-1)}{2}. \tag{E.56.16}$$

For the second formula, count a symplectic basis: its successive choices number $(p^4 - 1), p^3, (p^2 - 1), p$, and then divide by the two scalar matrices. The first formula is the same basis count in dimension two. Equality with $96!/2$ would imply $p \leq 96$, since p divides that order. Both displayed orders would then be smaller than $96^{10}/2$, while $96! \geq 77^{20} > 96^{10}$. This proves the asserted exclusions.

It follows from E.56.15 that no finite matrix target in the statement has an A_{96} quotient. A simple quotient of a product comes from one of its factors, as above. The old target has no such quotient either: its solvable kernel is killed in a simple nonabelian quotient, and the order of T_L has no factor eleven. The actual four- and five-strand Hurwitz maps are onto A_{96} by E.46. Goursat now proves exactly E.56.5 and E.56.6 at every finite modulus.

Every finite collection of congruence levels is dominated by their least common multiple. The closed actual images over R therefore surject onto every finite projection of the displayed products. Compactness proves the same product identities over R.

Finally E.55 retains the same Galois word in all matrix and T_L components and in the Hurwitz component. Their corrected maps are the displayed original joint maps precomposed

with one actual braid automorphism, so their images are the same products. The separate corrected T_{23} map is onto and agrees in S_L by E.55. The maximum solvable quotient of T_L is the specified S_L; the new matrix and alternating factors are perfect. Thus the same solvable-kernel Goursat argument gives the whole corrected product E.56.5. All inverse-limit comparisons retain these same maps. This proves the final assertion. □

The new certificate has the shared prefix `bc_symplectic_integral_targets`. It verifies the polynomial identities in E.56.9–E.56.10, the exact transpose and mirror comparison, and all 12096 saved level-five and level-seven permutation entries. It eliminates all $624 + 2400$ nonzero vectors over those two fields by literal original transvection words. It also constructs independent symplectic bases and factors 16 matrices at each of the moduli $5, 7, 25, 49, 35, 175, 245$, retaining a complete word witness at each modulus. Its run takes 3.076 seconds. Every protected earlier certificate is unchanged.